Problem Statement

To qualify the electronics hardware, ground control station, real-time data transmission, airborne telemetry, command system, and recovery system of a 25 kg re-entry capsule, a controlled drop test shall be conducted from an altitude of 5 km, within a 5 km operational radius.

The capsule shall be released into an atmospheric environment with horizontal wind speeds of 30–35 m/s, and all onboard sensing, avionics sequencing, telemetry, and pyrotechnic systems shall function correctly throughout the descent until recovery.

The purpose of this test is to measure, transmit, and log real-time environmental, structural, thermal, motion, and visual parameters during descent and to ensure successful parachute deployment, command reception, and recovery.

Requirements

1. Atmospheric Parameters

The capsule shall measure atmospheric parameters both outside and inside the vehicle during descent, including:

- Ambient pressure
- Ambient temperature
- Atmospheric density
- Molecular composition
- Humidity (outside the vehicle)
- Humidity (inside the vehicle)

2. Acceleration and G-Force

The capsule shall measure:

- G-force acting on the vehicle
- Acceleration using an accelerometer

3. Descent Velocity

The capsule shall measure and transmit:

- Real-time descent velocity

Position Tracking

The capsule shall provide:

- Real-time position of the vehicle during descent

CONFIDENTIAL

5. Vibration

The capsule shall measure:

- Vibration acting on the vehicle body

6. Surface Pressure

The capsule shall measure:

- Surface pressure at different locations of the vehicle

7. Surface Velocity

The capsule shall measure:

- Surface velocity at different locations of the vehicle

8. Thermal Parameters

The vehicle wall thickness is **7 cm**.

The capsule shall measure:

- Temperature at each 1 cm depth of the vehicle wall
- Total temperature data across the full wall thickness

9. Video

The capsule shall record:

- Upward-facing video
- Downward-facing video
- Side-view video

10. Wind Speed

The capsule shall measure:

- Horizontal wind speed

11. IMU

The capsule shall include:

- IMU data measurement during descent

CONFIDENTIAL

12. Drop Conditions

The re-entry capsule shall be dropped:

- From an altitude of 5 km
- With horizontal wind speed of 30–35 m/s

13. Parachute Deployment

The parachute shall:

- Open immediately after drop
- Deploy within less than 3 seconds

14. Avionics Sequencing

The capsule shall ensure:

- All avionics sequences function correctly
- All avionics real-time data is transmitted successfully

15. Pyrotechnic Devices

The capsule shall ensure:

- All pyrotechnic devices function correctly

16. Airborne Telemetry

The capsule shall ensure:

- Airborne telemetry system functions successfully throughout descent

17. Command

The capsule shall ensure:

- Command functionality operates correctly

CONFIDENTIAL